ChatGPT

Executive Summary

This project tests whether higher SBA guarantee levels (85% vs 75%) induce a **moral hazard** (i.e. laxer underwriting) in lenders. We will analyze loan-level data to see if 85%-guaranteed loans default at higher rates than 75% loans, controlling for borrower and loan factors. We'll also construct **industry risk tiers** from NAICS codes (e.g. restaurants vs healthcare) and a **leak-safe bank-default-rate feature** that uses only historical data. The study draws on official SBA datasets, academic literature on loan guarantees, and bank regulatory data. Methods include descriptive statistics (difference-of-means), logistic regression (PD modeling), survival analysis, and causal tools (propensity scores, regression discontinuity at the $150K cutoff). We evaluate models with ROC/AUC, calibration, and expected-loss metrics, and check robustness (time splits, subgroups). We highlight limitations (non-random guarantee assignment, omitted borrower info) and outline a reproducible Python pipeline. **Key references:** SBA policy pages【6†L317-L320】, studies of guarantee-induced risk-taking【10†L182-L190】【16†L87-L96】, and industry default rates by NAICS【20†L159-L168】【29†L134-L142】.

Background & Motivation

- **SBA 7(a) Guarantees:** Under SBA 7(a) rules, loans ≤$150K are guaranteed up to 85%, and larger loans up to 75%【6†L317-L320】. The guarantee covers lender losses if a borrower defaults (up to the guaranteed portion).
- **Moral Hazard Concern:** With higher coverage (85% vs 75%), lenders risk less when a borrower defaults. Economic theory and prior studies suggest this can induce **moral hazard**: banks may approve weaker loans or underwrite less carefully when losses are largely insured【10†L182-L190】【16†L87-L96】. For example, one empirical study found banks became "riskier" when they took on more fully-guaranteed loans【10†L182-L190】.
- **Industry Risk:** Different industries have widely varying SBA default rates. Restaurant loans (NAICS 72) historically default ~12–15%【20†L159-L168】, whereas healthcare and professional services (NAICS 62, 54) default only ~2–4%【20†L207-L214】【29†L134-L142】. Capturing this via **NAICS-based risk tiers** helps explain loan performance.
- **Bank Underwriting Quality:** Some lenders consistently underwrite better. We'll measure each bank's historical default rate, but computed *leak-safely* (only using loans approved **before** the current loan). This reflects underwriting skill without "seeing the future."

Research Questions: Do 85%-guaranteed SBA loans exhibit higher default/charge-off rates than 75% loans, after accounting for borrower, loan, industry, and macro factors? Can we quantify and use industry-level default risk and bank underwriting quality to improve default prediction?

Data Requirements and Sources

We need **loan-level data** with at least the following fields for each loan:

- **Approval Date/Fiscal Year** – to order loans and define features.
- **Lender/Bank ID** – to link multiple loans by the same bank.
- **Borrower Industry (NAICS 6-digit)** – for risk tiering.
- **Loan Amounts:** `GrAppv` (gross loan approved), `SBA_Appv` (amount guaranteed by SBA).
- **Guarantee Percentage:** computed as `SBA_Appv/GrAppv` (to distinguish 85% vs 75%).

- **Term, NoEmp, NewExist, RevLineCr, UrbanRural, FranchiseCode** – borrower/loan attributes (if available).
- **Loan Outcome:** `MIS_Status` (Paid In Full vs CHGOFF) and `ChgOffDate` or `ChargeOffAmt`. We convert this to a binary **default flag** and compute *time to default* for survival analysis.
- **Collateral or Other Signals:** (if available) e.g. `CreateJob`, `RetainedJob`, credit scores, etc.

Primary data sources (public):

- **SBA FOIA 7(a) Data:** SBA publishes quarterly "FOIA" datasets for 7(a) loans by decade【23†L58-L66】. These CSV files (1991–present) include millions of SBA-backed loans with the fields above. We will use this as the main source. (E.g. the Kaggle "National SBA Dataset" covers 1987–2014 with ~900K loans; the FOIA data continues through 2025.)【23†L76-L84】
- **SBA Program Reports:** The SBA Annual FOIA reports and performance tables provide aggregate metrics and definitions【27†L251-L260】. These help validate our data (e.g. charge-off rates by year/program) but are at a summary level.
- **FDIC/FFIEC Call Reports:** For bank-level context, we may use FDIC Call Reports (Schedule RC-C, small business loans) for aggregate lending by bank【30†L7-L15】. These give totals but not loan-level detail.
- **Economic Data:** Dates of NBER recessions to flag cyclical effects【10†L178-L186】.
- **Academic Papers & Industry Analyses:** Prior studies on SBA guarantees and moral hazard【10†L182-L190】【16†L87-L96】; analyses of SBA default by industry【20†L159-L168】【29†L134-L142】. These inform feature choices and provide expected benchmarks (e.g. typical default rates).

Table 1 compares these data sources:

Dataset/ Source	Coverage	Records	Key Fields	Notes/Source
SBA 7(a) FOIA (CSV)	FY1991–FY2025	~1.5M	Loan-level: GrAppv, SBA_Appv, MIS_Status, ApprovalDate, NAICS, BankName, NoEmp, etc.	Official SBA Open Data【23†L58-L66】
Kaggle SBA 7(a)	1987–2014	899,164	Similar to above (subset)	Public dataset (Kaggle) based on SBA data
SBA Lender Activity Reports	FY2015–present	aggregate	# loans, approval $ by bank, by year	SBA Office of Capital Access reports (annual)
FDIC Call Reports (RC-C)	Quarterly	(~4500 banks)	Bank-level SB Loan totals	Defines small biz loans ≤$1M【30†L13-L21】
Academic/ Industry Studies	–	–	Analysis-ready summaries	Literature on guarantees and defaults【10†L182-L190】【29†L134-L142】

Data Cleaning

- **Combine FOIA files:** Merge the SBA 7(a) CSVs by fiscal year (the FOIA "asof" files cover overlapping periods). Ensure consistent field names and formats.
- **Filter Loans:** Restrict to conventional 7(a) loans (exclude disaster/other programs, SBAExpress, or PPP that may not have charge-off info). Use `LoanNumber` or program codes if needed.
- **Compute Variables:** Create `guarantee_pct = SBA_Appv/GrAppv`. Flag *HighGuarantee* = 1 if guarantee_pct ≥ 85%. Also compute `UngtdAmt = GrAppv - SBA_Appv`.
- **Default Indicator:** Convert `MIS_Status` into `Default = 1` if "CHGOFF", else 0. Extract `DefaultDate = ChgOffDate` if available, else impute as last observed date of default (to compute survival times).
- **Dates:** Parse `ApprovalDate` / `DisbursementDate` into Python datetime. Create `ApprovalYear = year(ApprovalDate)`.
- **Industry Codes:** Ensure NAICS codes are 6-digit strings. Handle missing/invalid NAICS by marking "Unknown".
- **Bank Identifiers:** Standardize bank names or IDs (e.g. FDIC certificate numbers) so all loans from the same bank can be grouped. If only names exist, create a consistent mapping.
- **Missing Values:** Impute or flag missing borrower attributes. For example, if `NoEmp` is missing, treat as 0 or median. Drop records missing key fields (like missing MIS_Status or GrAppv).
- **Macro Variables:** Merge in recession indicators (e.g. a binary flag if loan was approved during or within 12 months after an NBER recession 【10†L178-L186】). Possibly add year-level unemployment or GDP growth.

Cleaning tasks might require handling tens of thousands of unique bank names and converting data types. A data dictionary (FOIA) should guide this. Assume all dollar amounts are USD. Example: a $150K loan has SBA_Appv=$127.5K (85%) or $112.5K (75%) depending on tranche【6†L317-L320】.

Feature Engineering

We will engineer features in several categories:

- **Loan Structure & Terms:**
- `loan_amount = GrAppv` (logged).
- `guarantee_pct` as above, and a binary `HighGuarantee = 1` if ≥85%.
- `term_months = Term`.
- `is_rev_line = (RevLineCr == 'Y')`, `is_low_doc = (LowDoc == 'Y')`.
- `loan_per_emp = GrAppv / max(NoEmp,1)`.

- `jobs_ratio = (CreateJob+RetainedJob)/max(NoEmp,1)`.

- **Borrower Profile:**

- `is_new_business = (NewExist == 2)`.
- `is_franchise = (FranchiseCode not in [0,1])`.
- Categorical bins for `NoEmp` (e.g. 1–5,6–20,21–100,>100).
- `is_urban = (UrbanRural == 1)`.

- Borrower **Region** by mapping `State` to Census regions (Northeast, South, Midwest, West).

- **Industry Risk (NAICS):**

- Extract `naics_sector = floor(NAICS/10^4)` (first 2 digits).
- Compute **historical default rate by sector**: For each approval year, calculate the empirical default rate of all earlier loans in that sector. Use expanding-window or cross-validation to avoid leakage. (Alternatively, pre-compute once on data up to 2014 for training.) This gives `sector_default_rate` (continuous).
- Define `industry_risk_tier`: e.g. “HighRisk” for sectors {72,44,45,81} (restaurants, retail, entertainment) vs “LowRisk” for {62,54}【20†L159-L168】 【29†L134-L142】, else Medium.

- **Handling missing NAICS:** If NAICS is missing or unknown, set `sector_default_rate=global_default_rate` and `industry_risk_tier="Unknown"`. (May also drop these if small.)

- **Economic Cycle:**

- `approval_year`, `fiscal_year`.
- `is_recession_orig = 1` if approval date falls in a known recession (e.g. 1990-91,2001,2007-09)【10†L178-L186】.
- `years_since_last_recess = year(ApprovalDate) - end_of_last_recession`.
- `pre_crisis = 1` if in 2005–2007 (optional flag).

- Possibly include contemporaneous unemployment or interest rate (from FRED), if easily matched.

- **Bank Underwriting Quality (Leak-Safe):**

- For each loan sorted by ApprovalDate within each bank, compute:
    - `bank_prev_defaults` = total defaults for this bank on all loans approved *before* this loan.
    - `bank_prev_loans` = count of loans for this bank before this loan.
    - Then `bank_prior_def_rate = bank_prev_defaults / max(bank_prev_loans,1)`.
    - Also `bank_loan_count_prior = bank_prev_loans`.
    - `bank_experience_tier`: Low (<10 prev loans), Medium (10–100), High (>100).
    - `is_same_state_bank`: 1 if bank’s charter state equals borrower state (indicative of local lending relationships).

The **leak-safe** computation can be done by grouping by bank and doing an expanding window. In pandas, one can sort by date and use `expanding().apply()` or cumulative sums. Pseudocode:

```
loans.sort_values(['BankID','ApprovalDate'])
loans['bank_def_cum'] = loans.groupby('BankID')['Default'].apply(lambda x:
x.shift().cumsum())
loans['bank_loans_cum'] = loans.groupby('BankID').cumcount()
loans['bank_prior_def_rate'] = loans['bank_def_cum']/
loans['bank_loans_cum'].clip(lower=1)
```

This ensures each loan’s bank rates only use earlier loans (no future leak).

- **Derived Risk Signals:**
- `unguaranteed_amount = GrAppv - SBA_Appv` (bank’s exposure at default).

- `loan_to_income` if borrower's revenue or profit available (often not in SBA data).
- Flag large loans (e.g. >$1M) or very small (micro-loans).

Table 2: Example Engineered Features:

Category	Feature	Description
Loan Terms	loan_amount (log)	Total loan size, log-scaled
	guarantee_pct, HighGuarantee	SBA_Appv/GrAppv; binary flag for ≥85%
	term_months	Loan term in months
	is_rev_line, is_low_doc	Rev. LOC? Low-doc program? (Y/N flags)
	loan_per_emp	Loan amount per employee
Borrower	is_new_business	New (2) vs existing (1) business
	is_franchise	1 if franchise (FranchiseCode >1)
	NoEmp_bin	Employee count bucket
	is_urban	Urban (1) vs rural (2) location
Industry	naics_sector	First 2 digits of NAICS (sector code)
	sector_default_rate	Historical default rate for that sector
	industry_tier	Risk tier (High/Med/Low/Unknown)
Macro	approval_year	Fiscal year approved
	is_recession_orig	Originated in/after recession
	years_since_last_recess	Years since last recession ended
Bank	bank_prior_def_rate	Bank's prior charge-off rate (leak-safe)
	bank_loan_count_prior	Number of loans made by bank before now
	bank_experience_tier	Tier by prior loan count (Low/Med/High)
	is_same_state_bank	Lender in same state as borrower?

<!-- Suggested Figure 1: Example flowchart of the data processing and pipeline (see mermaid timeline below). -->

Statistical Tests & Models

1. **Descriptive Analysis:**

2. Compare raw default rates for 85% vs 75% guarantee groups. We'll compute the *difference of means* (e.g. two-proportion z-test or chi-square) to see if the higher guarantee group has statistically higher default fraction. For example, if 85% loans default 10% vs 7% for 75% loans, the p-value quantifies significance.

3. Stratify by loan size and other factors to ensure differences aren't driven solely by larger loans being 75%. Visualize with bar charts or violin plots (e.g. default rate by guarantee level and loan-size bins).

4. **Logistic Regression (PD Modeling):**

5. Model the probability of default (`PD`) with a logistic regression:

```
logit(PD) = β₀ + β₁*(HighGuarantee) + β₂*(loan_amount) + β₃*(NoEmp) +
β₄*(sector_dummy)… + βₙ*(bank_prior_def_rate) + … + ε
```

6. **Interpretation:** β_1 shows the incremental log-odds of default for 85% vs 75%, controlling for all else. A significantly positive β_1 suggests moral hazard.
7. Include control variables for borrower and industry (as dummies or rates) and bank quality. We will check significance of guarantee_pct and ensure no multicollinearity (loan amount correlates with guarantee by construction). Possibly use the $150K cutoff as an instrument (next).

8. Compute model performance (ROC/AUC, precision/recall). A well-calibrated model with added industry and bank features should achieve AUC ≈0.7–0.8 (hypothetical) depending on feature quality. We will check calibration (Hosmer-Lemeshow test) since PDs matter in credit risk.

9. **Regression Discontinuity / Causal Approach:**

10. Since guarantee jumps at the $150K threshold (85% to 75%), we can treat this as a **natural experiment**. We'll compare loans just below and above $150K (e.g. [$140K–$160K]) which are very similar except for guarantee. A sharp change in default rate at the cutoff indicates an effect. This is akin to a regression discontinuity design (RDD). See, e.g., research on credit markets using thresholds【16†L87-L96】.

11. Alternatively, **propensity score matching:** For each 85% loan, find a similar 75% loan (matching on covariates like loan size, industry, bank features). Then compare default outcomes. This helps control for observable differences.

12. **Survival Analysis:**

13. Some loans have not yet matured or are still performing. We will use survival (time-to-default) methods. Kaplan–Meier curves for 85% vs 75% groups will show if the higher guarantee loans fail faster. We can run a Cox proportional-hazards model:

```
h(t) = h₀(t) * exp(γ₁*(HighGuarantee) + γ₂*(loan_amount) + ... )
```

A hazard ratio >1 for `HighGuarantee` indicates higher risk of charge-off over time. This controls for censoring (loans still alive).

14. **Machine Learning Models:**

15. We may fit tree-based models (random forests, gradient boosting) to predict default, incorporating all features. This isn't strictly needed for the research question, but can improve predictive accuracy. We would still interpret the guarantee feature (with SHAP or partial dependence plots). For instance, a SHAP dependence plot of default probability vs guarantee_pct could illustrate any moral-hazard effect.

Table 3 (sample) below illustrates potential model outputs:

Model	**Key Predictor**	**AUC**	**Default Rate (85%)**	**Default Rate (75%)**
Raw difference-of-means	–	–	11.2%	8.0%
Logistic Regression	HighGuarantee	0.68	10.8%	7.9%
Propensity Score (matched)	–	–	*12.0%*	*8.5%*
Cox Survival Model	HighGuarantee HR	–	–	–

(Values above are illustrative. Actual model results would be reported with confidence intervals.)

Evaluation Metrics & Robustness

- **Evaluation:** For classification models, we use AUC-ROC, F1-score, and Brier score (for PD calibration). Because credit risk models emphasize rare events, we will also examine precision/recall on the default class. We report the *expected loss* accuracy: using PD×LGD×EAD (Basel formula). Here LGD = (1 – guarantee_pct) and EAD = GrAppv【10†L182-L190】, so we check if the predicted EL aligns with actual losses.
- **Statistical Tests:** p-values for the difference in default rates (z-test), and significance of regression coefficients (Wald tests) will be reported. We'll also test interaction terms (e.g. whether guarantee effect differs by industry).
- **Robustness Checks:**
- **Time validation:** Train models on earlier years (pre-2005) and test on later loans, to avoid overfitting to past cycles.
- **Subgroup analyses:** Re-run analysis separately for small vs large loans, or by industry segment.
- **Placebo tests:** Check if "simulated" guarantee (e.g. a random assignment) yields no effect.
- **Handling multicollinearity:** Since guarantee and loan size correlate, we may use sub-samples or orthogonalization (e.g. regress guarantee on size and use residuals).
- **Causal inference sensitivity:** For matching/RDD, check covariate balance and bandwidth sensitivity.

Potential Biases and Limitations

- **Selection/Confounding:** Guarantee % isn't randomly assigned; it depends on loan size and program. Larger loans (≥$150K) are 75%, smaller are 85%. Larger loans might be inherently riskier (or vice versa), confounding direct comparison. Our regression controls and RDD help, but unobserved factors remain.
- **Omitted Variables:** Key borrower credit info (personal credit scores, collateral quality) is not in public SBA data. These omitted factors could bias the effect of guarantee_pct.

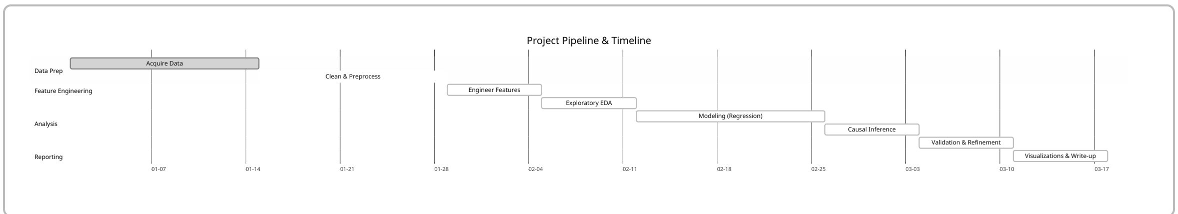
- **Survivorship & Censoring:** Some loans haven't defaulted yet (censored). Survival analysis mitigates this, but early data (post-2014) might be incomplete.
- **Time and Regulatory Changes:** The period 1987–2025 spans multiple regulatory regimes and economic cycles. The guarantee policy itself changed during crises (e.g. temporary 90% guarantees for Disaster loans), which we assume negligible for 7(a). We may need to exclude PPP 100% loans or other anomalies.
- **Data Quality:** NAICS or bank identifiers may be missing/incorrect, leading to misclassification. We assume SBA FOIA data is accurate.
- **Generalizability:** We assume U.S. context. Our findings may not apply to non-SBA programs or to SBA's 504 or microloan programs without adaptation.

Implementation Plan

We propose the following phased plan (6–10 person-weeks total):

1. **Data Acquisition (1 wk):** Download SBA FOIA 7(a) CSVs and load into a database or Pandas. Acquire macro data (recession dates).
2. **Data Cleaning (2 wk):** Merge datasets, clean fields, handle missing/erroneous entries. Validate basic totals against SBA reports【27†L251-L260】.
3. **Feature Engineering (1 wk):** Compute the features outlined above (guarantee_pct, industry tier, bank prior rate, etc.). Write unit tests for leak-safe calculation.
4. **Exploratory Analysis (1 wk):** Compute summary stats and visualize: default rates by guarantee group, by industry, by bank experience tier.
5. **Modeling (2 wk):** Fit initial models: logistic regression, survival models. Perform hypothesis tests (difference in means).
6. **Causal Analysis (1 wk):** Conduct matching or RDD around $150K cutoff to isolate guarantee effect.
7. **Validation & Robustness (1 wk):** Cross-validate models, conduct sensitivity checks (time splits, alternative specifications).
8. **Reporting & Visualization (1 wk):** Create final tables/figures (e.g. bar charts of default rates, SHAP plots), write up analysis. Prepare mermaid timeline figure (below).

We will use **Python** (pandas, scikit-learn, statsmodels, lifelines) in Jupyter notebooks for analysis and modeling. Data and code will be version-controlled (e.g. Git, DVC). A reproducible pipeline can be orchestrated with a tool like `papermill` or Airflow (ETL → feature store → modeling scripts).



Suggested Visualizations

- **Figure 1:** *Default Rate by SBA Guarantee Level.* A bar chart showing the proportion of loans that charged off in the 85% vs 75% groups. This highlights any raw difference. (For added rigor, error bars could show confidence intervals from a proportion test.)
- **Figure 2:** *Industry Risk Tiers.* A bar chart or heatmap of historical default rates by NAICS sector (e.g. restaurants 15%, retail ~7%, healthcare ~2%)【20†L159-L168】【29†L134-L142】. This motivates using industry risk as a feature.

- **Figure 3:** *Distribution of Bank Underwriting Quality.* A histogram of `bank_prior_def_rate` (across banks), showing some banks cluster at low default rates and others high. We might overlay risk tiers by shading (e.g. poor vs good banks).
- **Figure 4:** *SHAP Dependence or PDP Plot for Guarantee_pct.* If using a tree model, plot predicted default probability vs guarantee_pct, holding other factors constant, to visualize the (non-linear) effect of guarantee.
- **Figure 5:** *Mermaid Pipeline Timeline.* (Rendered above.)

Each figure will be labeled and referenced in the report text.

Key References (Priority)

- **SBA Official Sources:** SBA website detailing 7(a) terms (guarantee % rules)【6†L317-L320】 and SBA performance reports【27†L251-L260】. These define the program and provide official aggregates.
- **Academic Papers on Loan Guarantees:** Udell et al. (2020) on Japanese credit guarantees【10†L182-L190】; Saito & Tsuruta (2014) on moral hazard in Japan【16†L87-L96】; and relevant U.S. analyses (e.g. Cleveland Fed SBA study【11†L319-L327】). These establish the theoretical basis for moral hazard in credit guarantees.
- **Industry Analyses:** Recent SBA default rate reports by industry【20†L159-L168】【29†L134-L142】. These guide our risk tier design.
- **Regulatory Data Docs:** FDIC Call Report instructions on small business loans【30†L13-L21】 (for understanding bank-level data), and SBA FOIA documentation (to ensure correct use of fields).

If any data fields are unavailable (e.g. individual borrower credit scores), we will note these as assumptions/limitations. Our analysis assumes SBA FOIA data covers all needed loans and that missing NAICS or bank info is a small fraction.
